Assessing skin, hair, and nails

CHAPTER 14

WEBER

NURS3120: HEALTH ASSESSMENT AND PROMOTION

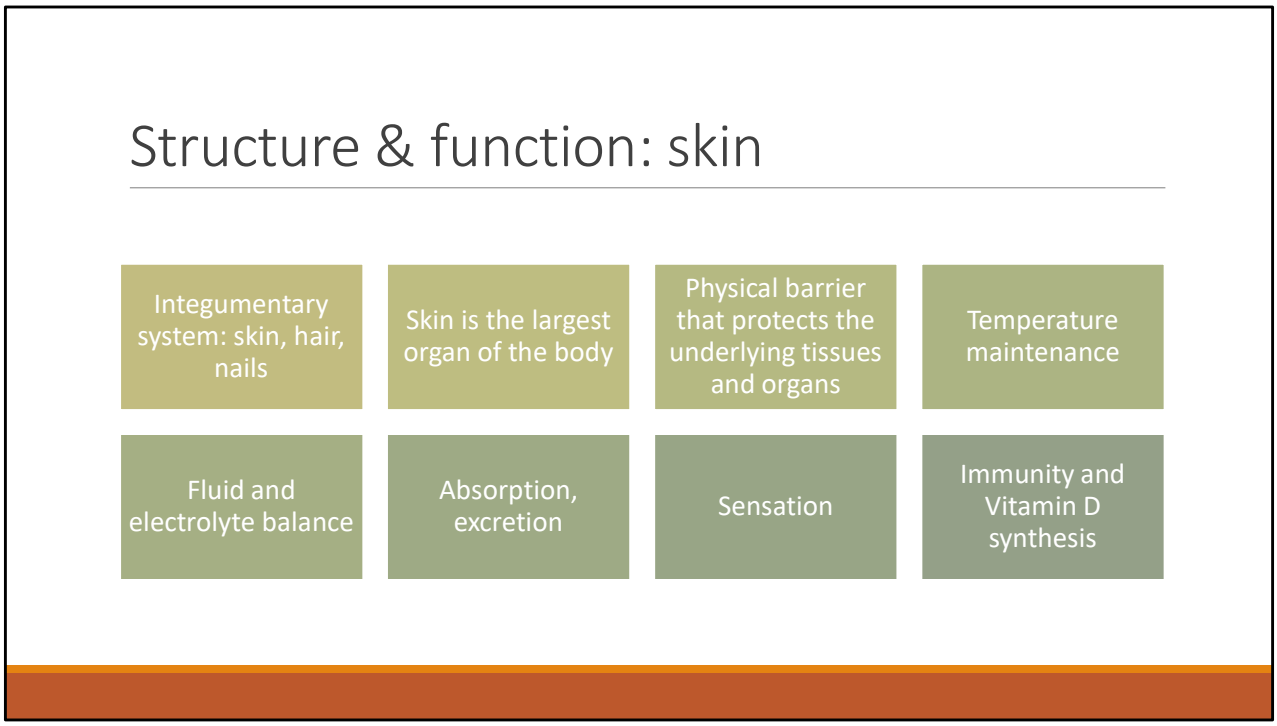
Objectives

1. Review the structure and functions of the skin, hair, and nails

2. Differentiate between normal and abnormal findings of the skin, hair, and nails

3. Analyze the data from the interview and physical assessment of the skin, hair, and nails to formulate valid clinical judgments

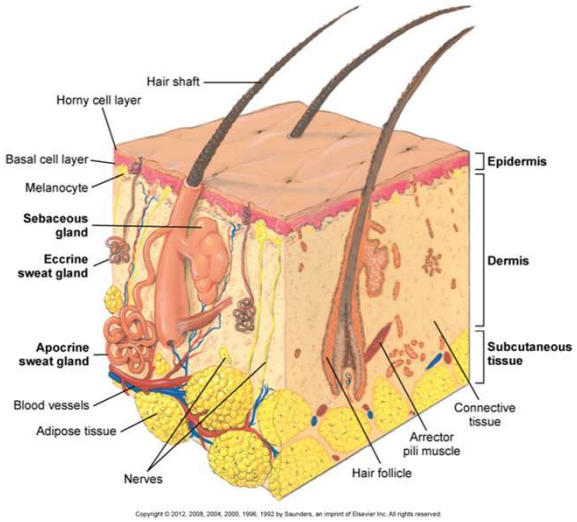
Structure & function: skin



The skin is the largest organ of the body. It is a physical barrier that protects the underlying tissues and organs from microorganisms, physical trauma, ultraviolet radiation (UVR), and dehydration. It plays a vital role in temperature maintenance, fluid and electrolyte balance, absorption, excretion, sensation, immunity, and vitamin D synthesis. The skin also provides individual identity to a person's appearance.

Perception – skin is a sensory organ with which able to feel pain, temperature, pressure and the feeling of someone touching .
Temperature regulation – skin allows for heat dissipation through the sweat glans and heat storage through subcutaneous insulation
Identification – We identify others through facial characteristics such as hair and skin color and our fingerprints.
Communication – Emotions are expressed with skin such as blushing or blanching as a sign of distress
Wound repair – cell replacement or wound healing of surface wounds
Absorption and excretion – excretion though our sweating of by –products of cellular composition (uric acid)
Production of vit D – skin is the surface in which UV light converts cholesterol into

vitamin D.



Layers of the skin

Overview- largest single organ of the human body. The dermis is a well-vascularized, connective tissue layer containing collagen, elastic fibers, nerve endings, and lymph vessels. It is also the origin of sebaceous glands, sweat glands and hair follicles.

Takeaway: The Epidermis is the outer layer of skin. It is the first layer and that is important when we talk about pressure injuries. The dermis and subcutaneous tissue also varies on the person, some people have more than others. Also look at where the adipose tissue is located as well, this will become important once again with the aging client and with pressure injuries.

Structure & function: hair

VELLUS

- "Peach fuzz"
- Present over most of the body
- Provides thermoregulation by wicking sweat away from the body
- Shorter, pale, and fine

TERMINAL

- Particularly scalp and eyebrows
- Longer, coarser, and generally darker
- Initiated additional growth by puberty (Axillae, perineum, and legs)
- Color varies and is determined by the type and amount of pigment production

The hair consists of layers of keratinized cells, found over much of the body except for the lips, nipples, soles of the feet, palms of the hands, labia minora, and penis. Hair growth occurs at the base of the follicle. There are two types of general hair: vellus and terminal.



Structure & function: nails

- Hard, transparent plates of keratinized epidermal cells
- Nail body is pink-tinged from blood vessels underneath
- Lunula: Crescent shaped area at the base of the nail
- Protect the distal ends of the fingers & toes
- Enhance precise movements of the digits
- Allow for an extended precision grip

Nails grow from the cuticle. This is a normal picture of what a nail should look like. If your nails do not look similar to this picture, it could mean an underlying systemic issue. The lunula is crescent-shaped area located at the base of the nail.



Collecting subjective data

What questions would the nurse ask related to skin, hair and nails?

SKIN:
Past history of skin disease, allergies, hives, psoriasis, or eczema? Food allergies can precipitate a variety of skin eruptions
Change in pigmentation or color? Changes in pigmentation can lead to skin cancer.
Change in mole size, shape, color, tenderness?
Excessive dryness or moisture?
Pruritus or skin itching?
Excessive bruising? Just be observant if the excessive bruising does not match how they got it; it could be a sign of abuse.
Rash or lesions?
Medications: prescription and over-the-counter? Medications can cause a photosensitivity reaction if the skin is exposed to UV light.
Environmental or occupational hazards? Are they a potential irritate or damage to the skin
Self-care behaviors? Do you use sunscreen or a SPF.
Sunbather/suntanner? Use sunscreen? Not using sunscreen puts a client at risk for skin cancer.
Tattoos- 5 types, traumatic, amateur, professional, medical (radiation markings), cosmetic

HAIR:
Hair loss? Some medications can be a cause of hair loss, but hair loss is common with aging because the rate of hair growth slows, and hair strands become thinner. Chemotherapy clients can lose their hair. Malabsorption syndromes, malnutrition, anorexia nervosa and bulimia
Change in nails' shape, color, or brittleness? Nail changes may be seen in systemic disorders such as malnutrition or with local irritation, Bacterial infections/fungal infections

Inspection: skin

Color
- General pigmentation, freckles, moles, birthmarks
- Note color variations: Vitiligo, pallor, cyanosis, jaundice, acanthosis, erythema

Integrity
- Skin should be intact without any reddened areas

Lesions
- Skin should be smooth without lesions

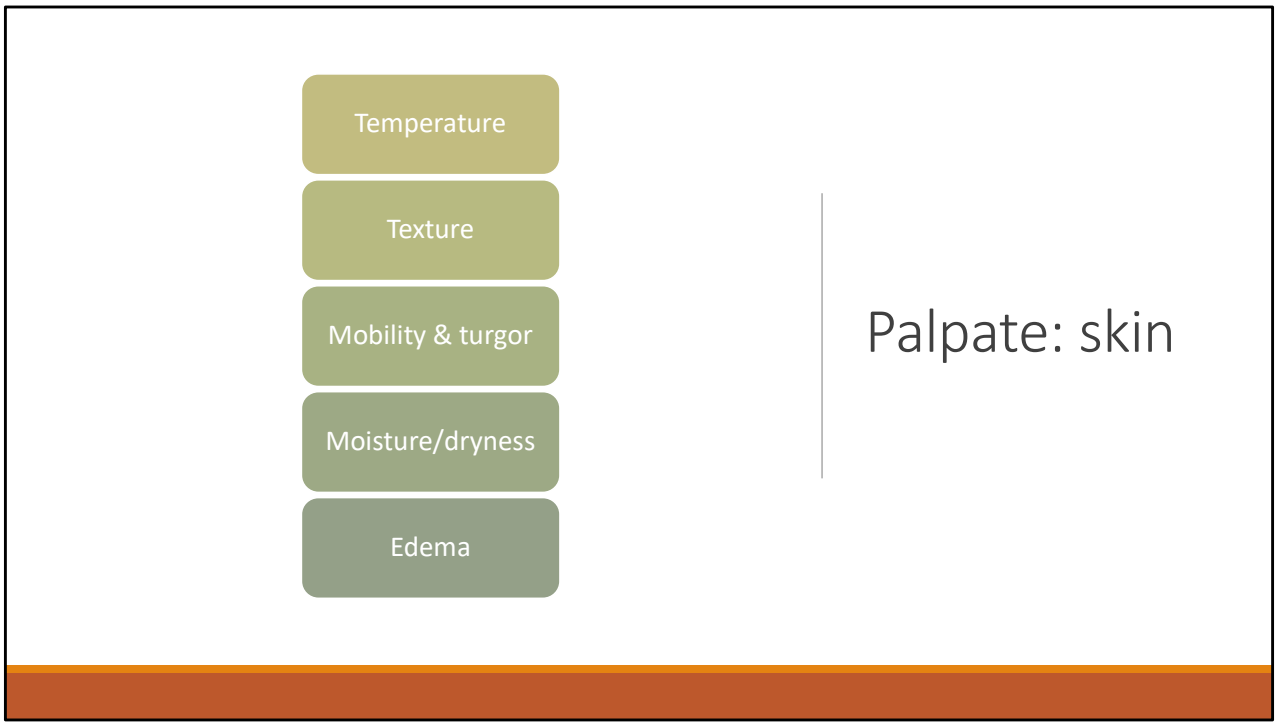




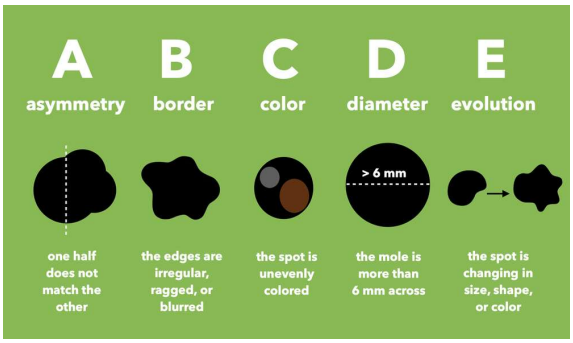
Inspection: Normal: evenly colored skin tones without unusual or prominent discoloration
Vascularity or bruising: Multiple bruises at different stages of healing and excessive bruises above knees or elbows should raise concern about physical
abuse needle marks or tracks from intravenous injection of street drugs may be visible on antecubital fossae, forearms, or on any available vein
These are abnormal:
Pallor (loss of color)
Color – **Vitiligo** – absence of melanin or when pigment producing cells die or stop functioning
Cyanosis indicates hypoxemia – as seen in asthmatics, chronic bronchitis, cardiac arrest, heart failure
Jaundice as with hepatitis, cirrhosis, sickle-cell disease, a transfusion reaction
Acanthosis nigricans is a velvety darkening of skin in body folds and creases especially the neck, groin, and axilla
Erythema: reddened areas on the skin
Assess skin integrity: Pay attention to pressure point areas, skin breakdown is initially noted as reddened area on the skin that may progress to serious and painful pressure ulcers

Lesions: observe the skin surface to detect abnormalities. Skin should be smooth without lesions. Lesions may indicate local or systemic problems. Skin irritation to diseases. Vascular lesions, arterial lesions.

Bold words depicted on slide



Palpate the skin to assess texture: Use the palmar surface of your three middle fingers to palpate skin texture should be smooth: Abnormal is rough, flaky dry skin
Palpate for temperature: Use the dorsal surfaces of your hands to palpate the skin, skin should be warm, cold/cool skin is abnormal can be seen with a circulatory issue or if hot could be a fever.
Palpate for mobility and turgor: Ask the client to lie down. Using two fingers, gently pinch the skin over the clavicle. Normal, skin is mobile with elasticity and returns to original shape quickly. Decreased mobility is seen with edema. How easily can the skin be pinched? Turgor refers to the skin's elasticity and how quickly the skin returns to tits original shape after being pinched.
Palpate the skin for moisture or dryness: are they diaphoretic (sweating)?
Edema: Push down on the pretibial area to check for edema (swelling related to accumulation of fluid in the tissue) normal: skin rebounds and does not remain indented when pressure is released.



ABCDE assessment of lesions

A: Asymmetry

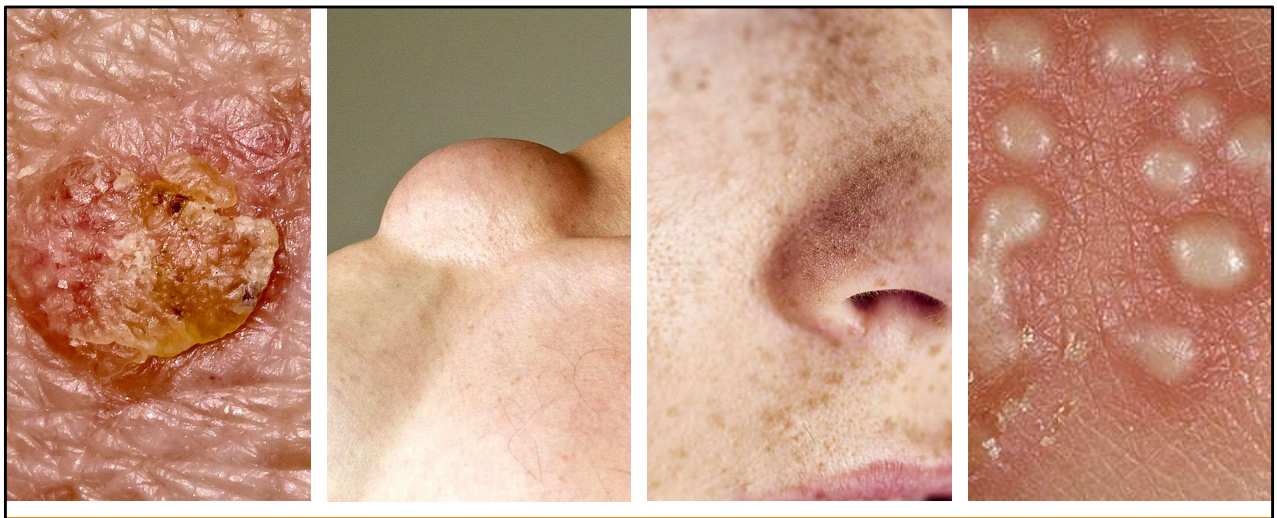
B: Border

C: Color

D: Diameter

E: Evolution (changing)/ elevation/ exudate (note color or odor)

We want to continue to monitor and assess these lesions and assess them for possible skin cancer. This mnemonic is used to detect signs of skin cancer that indicate the need for further medical evaluation.



Primary skin lesions: abnormal findings

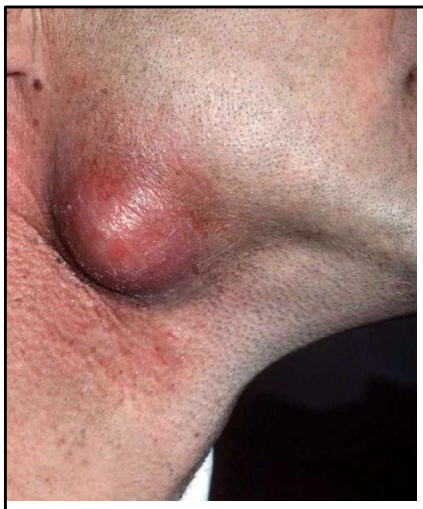
Papule/Plaque; Nodule/Tumor; Macule/Patch; Vesicle/Bulla

Primary skin lesions arise from normal skin due to irritation or disease.
Papule and plaque: elevated, palpable, solid mass: elevated nevi, warts; psoriasis or **actinic keratosis**

Nodule and tumor: elevated, solid, palpable mass that extends deeper into the dermis than a papule: keloids, **lipoma**, squamous cell carcinoma

Macule and patch: small, flat, nonpalpable skin color change macules are less than 1 cm with a circumscribed border whereas patches are greater than 1 cm and may have irregular border: Examples include **freckles**, flat moles, petechia, rubella, vitiligo, port wine stains, and ecchymosis

Vesicle and bulla: circumscribed elevated, palpable mass containing serous fluid: **herpes simplex zoster**, chicken pox, poison ivy, second degree burns
Bold words are depicted on slide





Primary skin lesions: abnormal findings

Cyst; Pustule; Wheal

Cyst: encapsulated fluid-filled or semisolid mass that is located in the subcutaneous tissue or dermis: **sebaceous cyst** or epidermoid cyst

Pustule: pus-filled vesicle or bulla: **acne**

Wheal: elevated mass with transient borders that is often irregular, does not contain free fluid: **hives** or insect bites.

Bold words are depicted on slide



Secondary skin lesions

Ulcer; Scar; Fissure; Erosion

Secondary skin lesions arise from changes in primary lesions.
Ulcer: Skin loss extending past epidermis: venous insufficiency or **pressure ulcers**

Scar: skin mark left after healing of wound or lesion that represents replacement by connective tissue: **surgical incisions**

Fissure: linear crack in the skin that may extend to the dermis and may be painful: chapped lips or **athlete's foot**.

Erosion: loss of superficial epidermis that does not extend to the dermis: canker sore (aphthous stomatitis), **scratch marks**

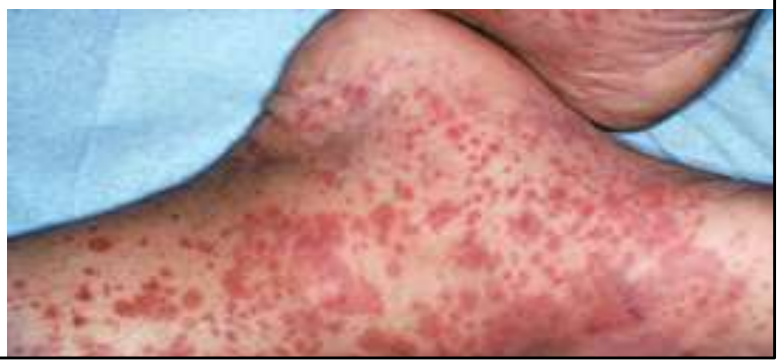
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Vascular skin lesions

Ecchymosis; Hematoma; Petechia







Vascular skin lesions are associated with bleeding, aging, circulatory conditions, diabetes, pregnancy, and hepatic disease, among other problems.
Ecchymosis: (bruise) color varies and changes: black, yellow, purple. It is secondary to blood extravasation and associated with trauma and bleeding tendencies.

Hematoma: collection of blood creating an elevated ecchymosis: associated with trauma

Petechia: round red or purple macule that is 1-2cm in size. It is secondary to blood extravasation and associated with bleeding tendencies or emboli to skin.

Vascular skin lesions

Spider angioma; Cherry angioma; Telangiectasis



Spider angioma: red arteriole lesion with a central body with radiating branches, common places face neck arms and trunk. It is associated with liver disease, pregnancy, and vitamin B deficiency.

Cherry angioma: papular and round, red or purple lesion found on the trunk or extremities. It may blanch with pressure. Normal age-related skin alteration.

Telangiectasis: bluish or red lesion with varying shape (spider like or linear) found on the legs and anterior chest. Does not blanch when pressure is applied. It is secondary to superficial dilation of venous vessels and capillaries and associated with increased venous pressure states (varicosities).





Inspect & palpate: hair

Wear gloves!

Color

Texture

Distribution

Lesions

Inspect the scalp and hair for general color and condition: Excessive scaliness may indicate dermatitis, raised lesion may indicate infections or tumor growth, dry hair may be seen with hypothyroidism

Inspect amount and distribution of scalp, body, axillae, and pubic hair. Look for unusual growth elsewhere. Excessive generalized hair loss may occur with infection, nutritional deficiencies, hormonal disorders, thyroid or live disease, drug toxicity, or renal failure. Chemo or radiation therapy.

Inspect for lesions

Dandruff; Alopecia

Inspect & palpate: nails

Inspect nails for:

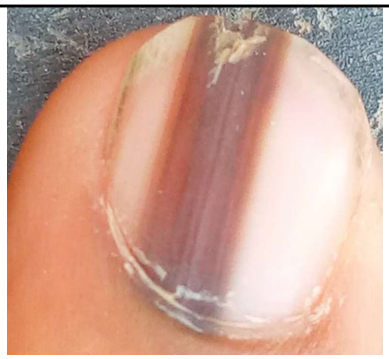
Grooming & cleanliness

Color & markings; should be pink tones

Shape; 160-degree angle between nail base and skin

Palpate for:

Capillary refill; Pink tone should return to nail in <2 seconds





Nail grooming: dirty, broken, or jagged fingernails may be seen with poor hygiene or a client's hobby or occupation.
Inspect nail color and markings: pale or cyanotic nails may indicate hypoxia or anemia. Splinter hemorrhages may be caused by trauma. **Beau lines** occur after acute illness and eventually grow out. Yellow discoloration may be seen with fungal infections or psoriasis. Nail pitting is also common in psoriasis. Color – brown streaks suddenly appearing could be **melanoma**; cyanotic nail beds are related to cardiovascular or respiratory dysfunction.
Shape: Clubbing is greater than 160 degrees, can occur from hypoxia. Spoon nails (concave) may be present with iron deficiency anemia.
Cap Refill: greater than 2 seconds may be seen with respiratory or cardiovascular diseases that cause hypoxia.
Bold words depicted on slide







Abnormal findings: nails

Onychomycosis; Paronychia; Clubbing

Onychomycosis: fungal infection of the nails, common in older adults, causes thickening, discoloration and changes in the nail integrity

Paronychia: inflammation indicates local infection

Clubbing: commonly associated with lung and heart diseases

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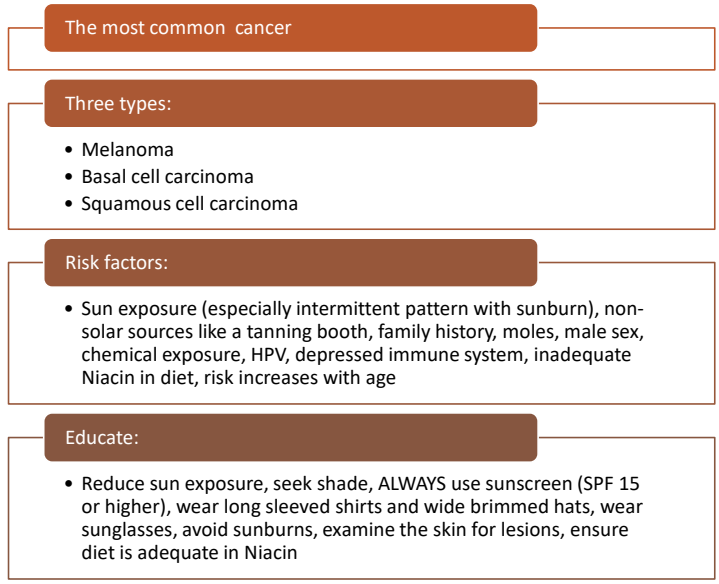


Abnormal findings: nails

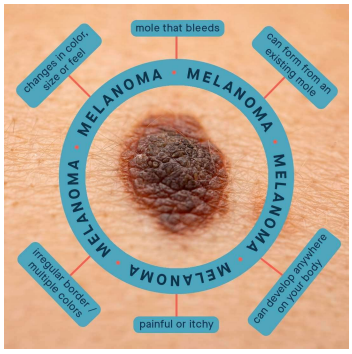
Yellow Nail Syndrome: grow slow and are curved . May be seen with AIDS and respiratory syndromes.

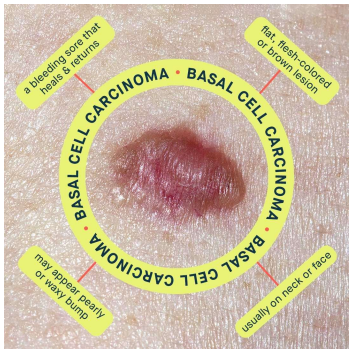
Koilonychia: spoon shaped nails maybe seen with trauma to cuticles or nail folds or in iron deficiency anemia, endocrine or cardiac disease.

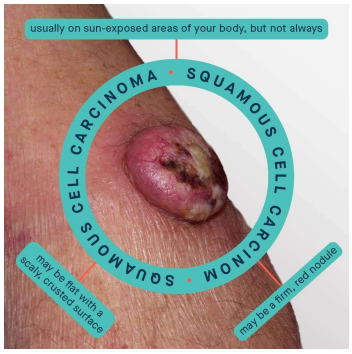
Evidence-Based health promotion and disease prevention: Skin cancer



One in five Americans will develop skin cancer by age 70
Cancerous lesions can be classified as either primary or secondary lesions
Niacin (Vitamin B3) sources:
Liver, chicken, tuna, turkey, salmon
Peanuts, avocado, brown rice, whole wheat

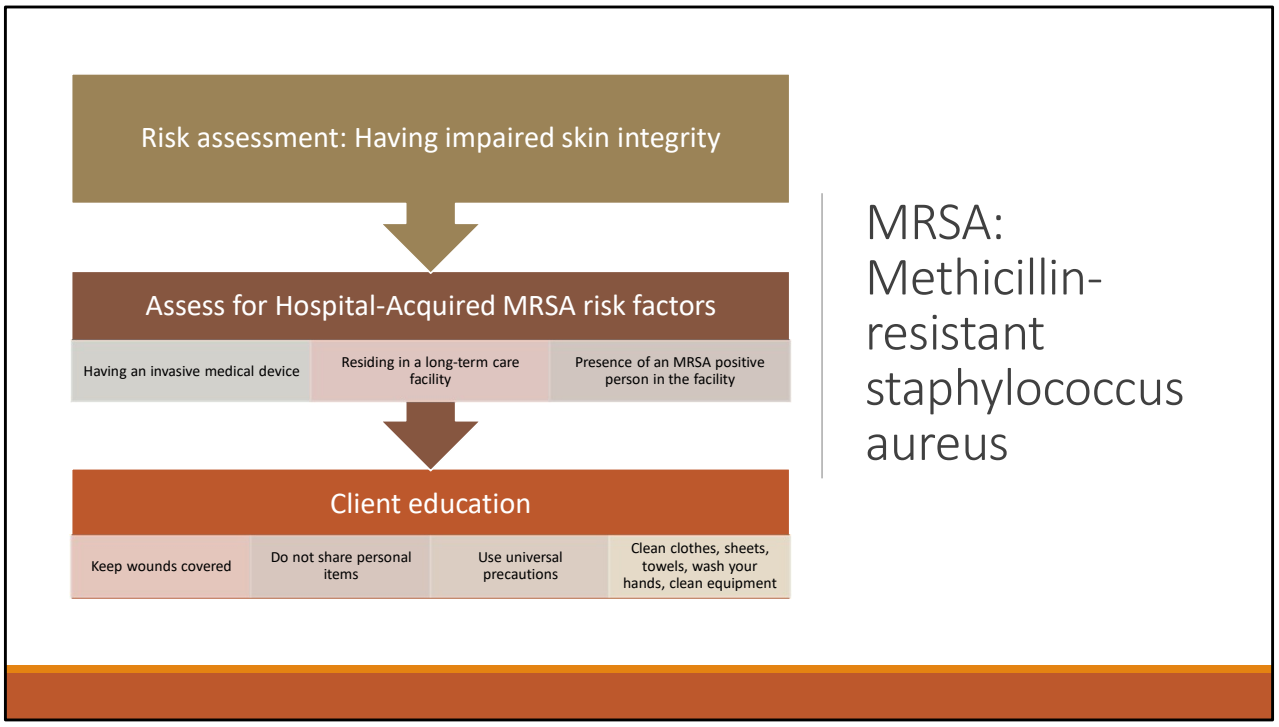






Skin cancer

Pictures from Dr. Pimple popper facebook page



SUPER BUG!!!

Methicillin-resistant Staphylococcus aureus (MRSA) infection is caused by a type of staph bacteria that's become resistant to many of the antibiotics used to treat ordinary staph infections.
Most MRSA infections occur in people who've been in hospitals or other health care settings, such as nursing homes and dialysis centers within the last year. When it occurs in these settings, it's known as hospital acquired MRSA (HAI). HAI-MRSA infections usually are associated with invasive procedures or devices, such as urinary catheters, surgeries, intravenous tubing or artificial joints. HA-MRSA can spread by health care workers touching people with unclean hands or people touching unclean surfaces.

The greatest risk factor for MRSA is impaired skin integrity

Staph skin infections, including MRSA, generally start as swollen, painful red bumps that might look like pimples or spider bites. The affected area might be:
•Warm to the touch
•Full of pus or other drainage

•Accompanied by a fever

These red bumps can quickly turn into deep, painful boils (abscesses) that require surgical draining. Sometimes the bacteria remain confined to the skin. But they can also burrow deep into the body, causing potentially life-threatening infections in bones, joints, surgical wounds, the bloodstream, heart valves and lungs.



Older adult considerations

Decreased flexibility and mobility may impair the ability to maintain proper hygiene such as bathing, cutting nails, and hair care

Skin turgor is less elastic, drier, and may easily tear; sagging, wrinkled skin

Loss of color pigmentation in hair; turns white & coarse

Nails can become either thick or thin and yellow

Perspiration decreases with aging because sweat gland activity decreases

Dry skin – skin itches and looks flaky and loose
Texture – acrochordons or skin tags
Skin turgor – skin will tent due to less elasticity the older client will also show more prominent facial bones because of the sagging skin or wrinkled skin, it will also affect their breasts and scrotal area.

Older adult considerations

Seborrheic keratosis; Purpura; Cutaneous horn; Senile lentigines

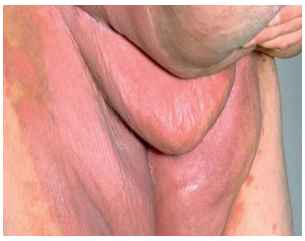








Senile Lentigines – common normality (liver spots) - melanocytes that appear after much sun exposure
Seborrheic Keratoses – raised, thickened areas of pigmentation that looks corseted, scaly and sometimes warty.
Purpura- Small blood vessels leak under the skin
Cutaneous horn- is a hard conical projection from the skin, made of compact keratin.


Moisture associated skin damage

Moisture Associated Skin Damage; Also known as MASD. This is skin breakdown associated with moisture. This is why this is very important to do a skin assessment. It can be found underneath the folds of skin, under the breasts, under the pannus, and on the buttocks.

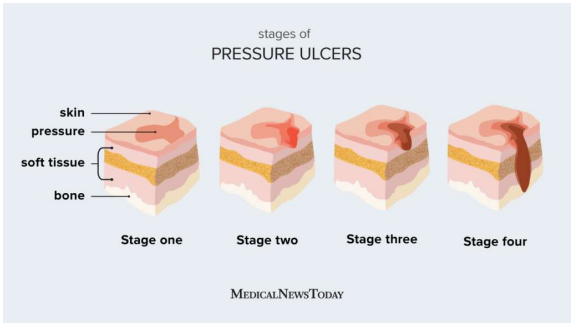
Braden scale

Assessment tool for risk of pressure injury

Assess:

- Sensory perception
- Moisture
- Activity
- Mobility
- Nutrition
- Friction and shear

Braden Scale widely used across the healthcare system to assess if a client is at risk for pressure injury:
Score 5-23. the lower the score the more at risk for developing a pressure injury



Evidence-Based health promotion and disease prevention: Pressure injuries

Major cause of morbidity and mortality

Costly to both client and institution

- In terms of both pain and suffering as well as dollars spent

Early assessment is key

Healthy People 2030: Reduce the rate of pressure injury related hospitalizations

Pressure injuries are a major cause of morbidity and mortality. The most significant contributing factor to pressure injury development is unrelieved pressure, but friction and shear also can contribute or worsen the condition. Prevalence of pressure injuries varies by bed type (higher quality foam and others) and clinical area, but occur more frequently in critical care, long-term care facilities, and in clients at high risk, such as those on prolonged bed rest.

Braden scale used for risk assessment



Pressure injuries

Client education:

- Bathe with mild soap, limit friction, warm not hot water
- Use moisturizers for dry skin and avoid low humidity and cold air
- Avoid vigorous massage and avoid massage over bony prominences
- Take nutritional supplementation as directed
- Use incontinence skin cleansing methods as needed

For bed or chair-bound clients:

- Reposition Q15 min for chair and Q2 hr for bed
- Use a pressure mattress or chair cushion
- Use lifting devices
- Use positioning devices



Pressure injury: stage 1

SKIN IS INTACT, RED, AND DOES NOT BLANCH WHEN PRESSURE IS APPLIED

Pressure injury used to be called decubitus ulcers. This is a huge problem in the hospital and nursing homes.
Occur over bony prominences but can tunnel to non- bony places.





Pressure injury: stage 2

Partial thickness skin erosion

First layer of skin is broken

Also includes an intact blister

Blisters can form from friction in the hospital beds, but be mindful that not all blisters are a result from a pressure injury.





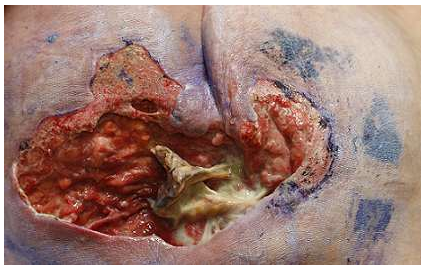
Pressure injury: stage 3

Full thickness ulcer extending into the subcutaneous tissue, crater like

Subcutaneous fat may be visible

Can tunnel down under the skin where not visible

Elbow
Base of spine



Pressure injury: stage 4

BONES, TENDONS, MUSCLE ARE VISIBLE

Pressure injury: unstageable

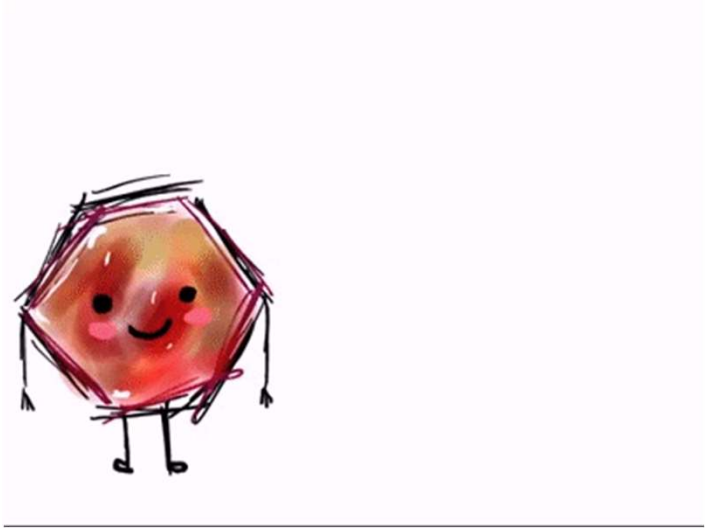
Unable to see the wound base

Think of an iceberg, you can only see a small representation of what is potentially underneath





Unstageable pressure injuries: Need to be evaluated. There are two types of wounds. One's that have are hard and appear to have developed a protective barrier around the wound or an unstageable wound that needs to be debridement. Unstageable tissue is usually black and necrotic. This can act as a protective barrier or needs to be surgical removed or pharmacologically removed. If the area around the wound bed is red/pink and irritated and the wound bed, if pushed upon is soupy and soggy and moveable, then underneath that layer is probably some type of infection, and it needs some assistance.



Questions?